

Deep Learning Guided Discovery and FEP Validation of NOD2 Binders for the R702W Crohn's Variant

Tanmay Vasudeva

Texas Virtual Academy at Hallsville, Texas, USA

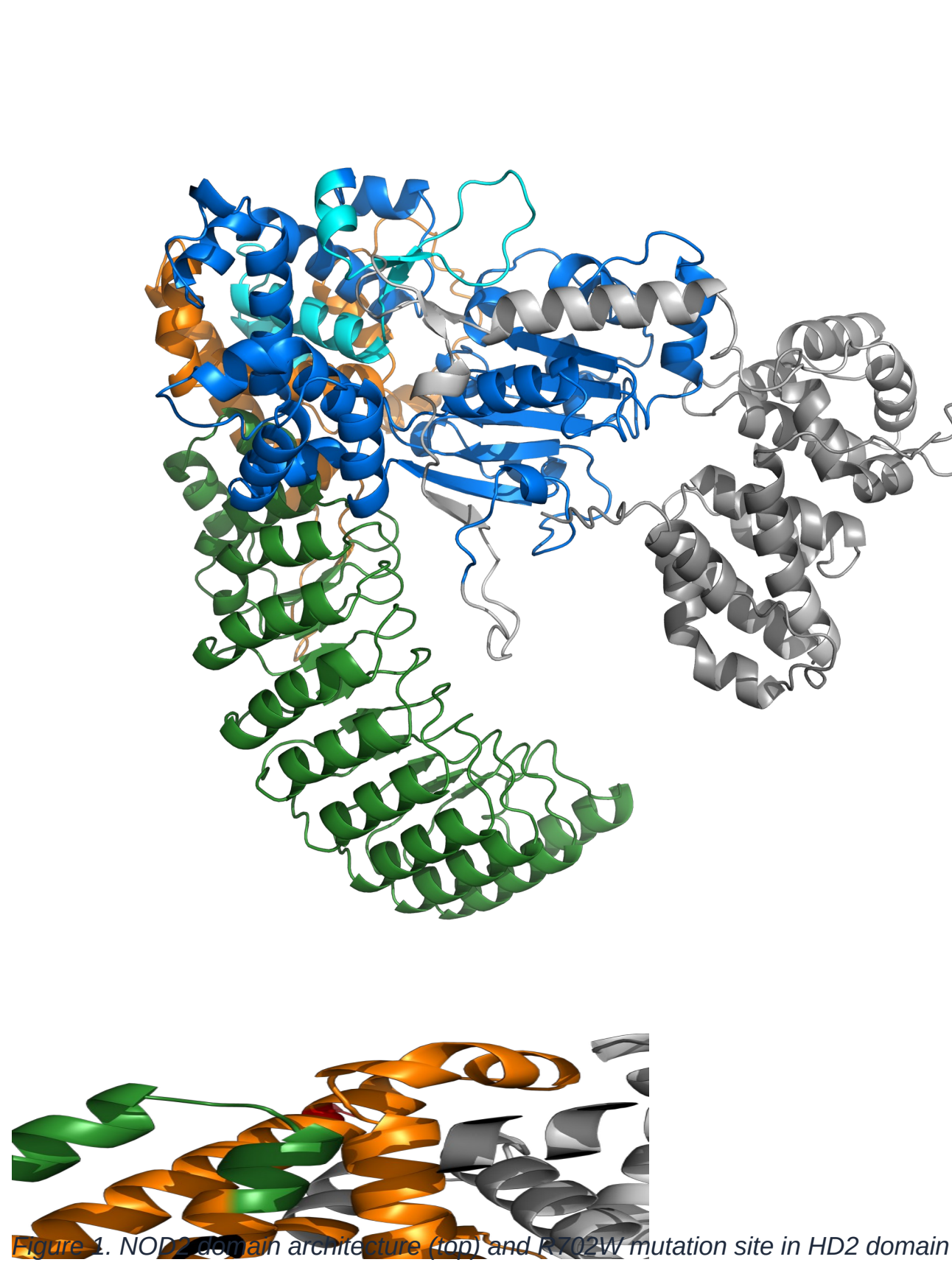
Purpose

Crohn's disease is a chronic inflammatory bowel condition affecting over 3 million people worldwide, with the NOD2 R702W variant conferring a 3-fold increased disease risk. Despite this strong genetic association, no FDA-approved therapeutics specifically target NOD2, and ligand binding to this receptor remains poorly characterized. The purpose of this study was to develop a computational pipeline integrating deep learning-based molecular docking with free energy perturbation (FEP) calculations to identify small molecules that bind NOD2's leucine-rich repeat (LRR) domain, and to determine whether the R702W mutation differentially affects binding affinity—potentially enabling mutation-specific therapeutic strategies.

Research Question

Can small molecules be identified that bind NOD2's LRR domain with differential affinity between wild-type and R702W mutant protein, and can such differential binding be quantified using rigorous free energy perturbation methods to enable mutation-specific therapeutic targeting?

Background



NOD2 (nucleotide-binding oligomerization domain-containing protein 2) is a cytoplasmic pattern recognition receptor that detects bacterial peptidoglycan fragments, triggering innate immune responses critical for intestinal homeostasis. The protein comprises four functional domains: two N-terminal CARD domains (gray) for downstream signaling, a central NBD domain (blue) for nucleotide binding and oligomerization, an HD2 helical domain (orange) containing the R702W mutation site, and a C-terminal LRR domain (green) responsible for ligand recognition.

The R702W mutation, located 79.4Å from the predicted binding pocket, represents a paradigm for allosteric regulation—the substitution of a charged arginine with a bulky tryptophan induces conformational changes that propagate through the protein structure to affect distant functional sites.

Methods and Procedures

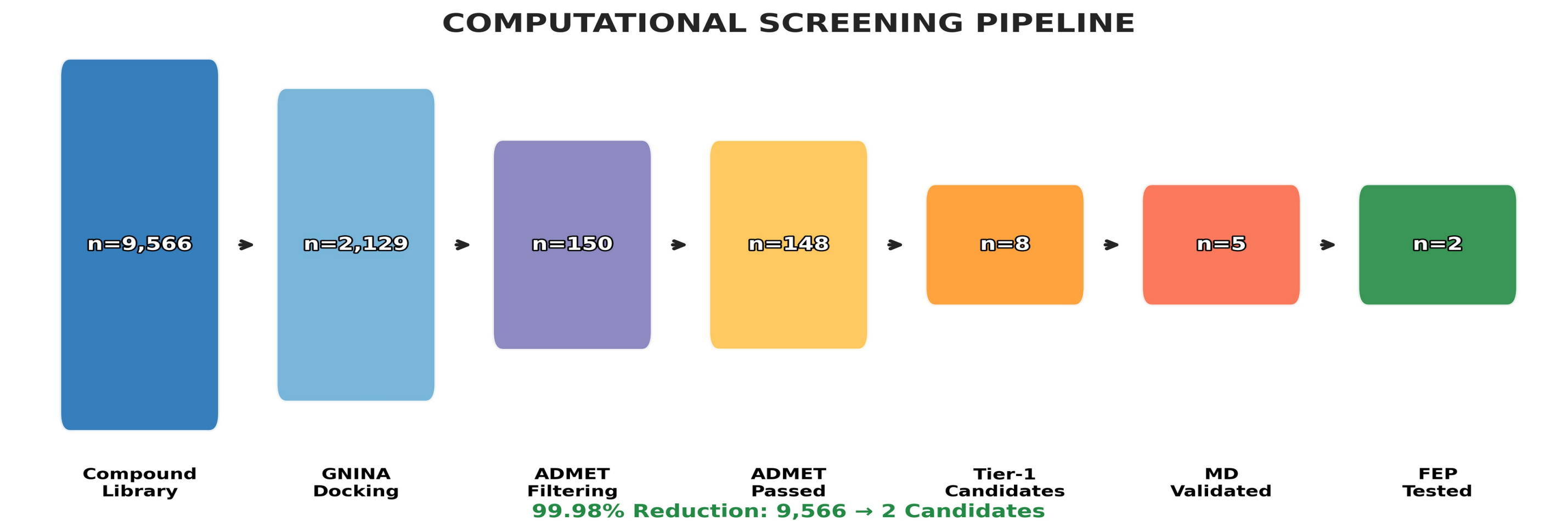


Figure 2. Computational screening pipeline reducing 9,566 compounds to 2 validated candidates.

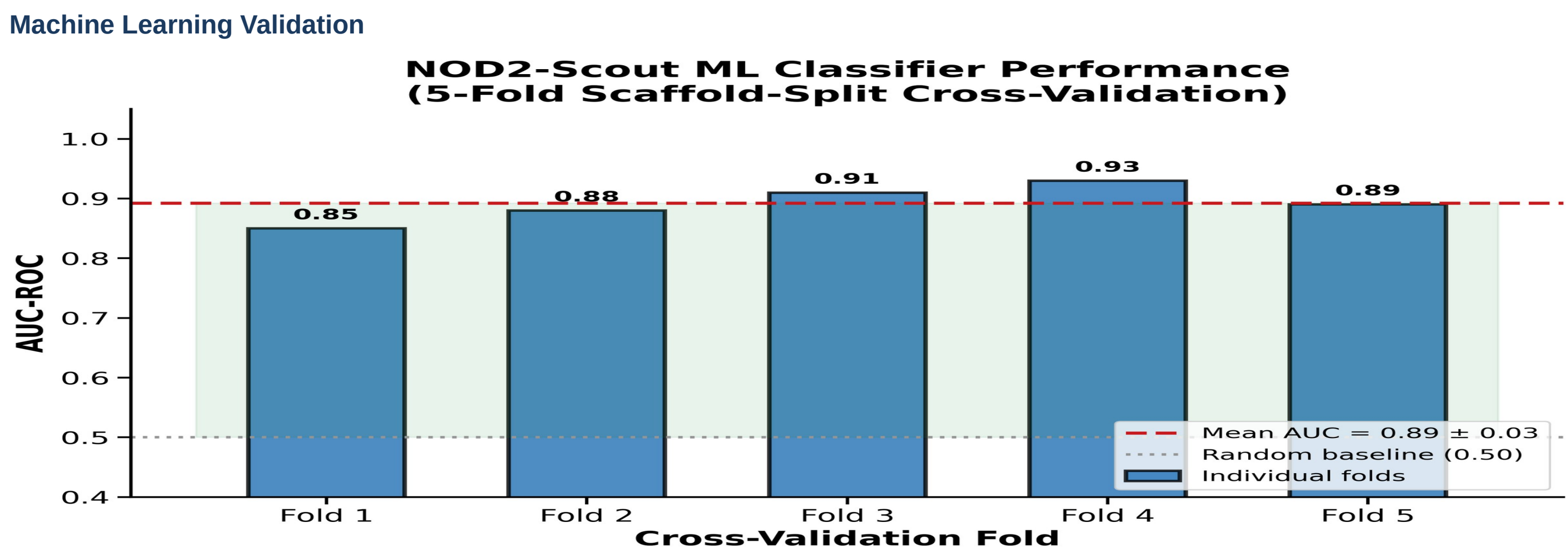
Virtual screening was performed on a library of 9,566 compounds comprising 2,152 FDA-approved drugs from e-Drug3D and 7,414 natural products from the COCONUT database. All compounds were docked against an AlphaFold2-predicted NOD2 structure using GNINA, a deep learning-enhanced molecular docking program that employs convolutional neural networks to score protein-ligand interactions.

To distinguish true binders from false positives, NOD2-Scout, an XGBoost gradient boosting classifier, was trained on 2048-bit Morgan fingerprints derived from the docking results. Model validation employed rigorous scaffold-split 5-fold cross-validation to prevent data leakage, with shuffled label controls confirming that the classifier learned genuine binding patterns rather than dataset artifacts.

Top-ranked candidates underwent ADMET (absorption, distribution, metabolism, excretion, toxicity) filtering using ADMETlab 2.0, followed by 520+ nanoseconds of molecular dynamics (MD) simulations in GROMACS to assess binding stability. Apo simulations of wild-type and R702W NOD2 characterized mutation-induced conformational changes.

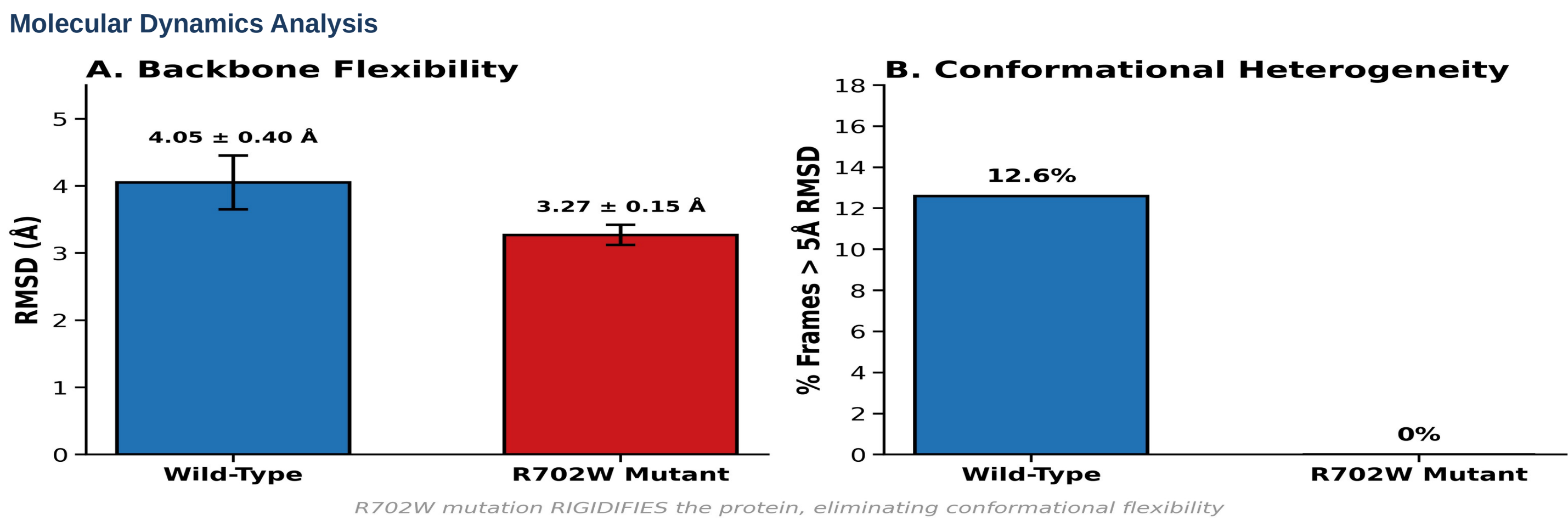
Free energy perturbation (FEP) calculations were performed using the pmx workflow with 20 Å windows and 5 ns equilibration per window to quantify binding free energy differences ($\Delta\Delta G$) between wild-type and mutant protein with statistical rigor.

Results



The NOD2-Scout classifier achieved ROC-AUC values of 0.85-0.93 across five scaffold-split cross-validation folds (Figure 3), with a mean AUC of 0.89 ± 0.03 . Critically, shuffled label controls yielded AUC values near 0.50, confirming that the model learned genuine structure-activity relationships rather than memorizing training data artifacts.

Figure 3. NOD2-Scout 5-fold scaffold-split cross-validation performance.



Apo molecular dynamics simulations revealed a striking difference between wild-type and R702W NOD2 conformational dynamics. Wild-type protein exhibited an average backbone RMSD of $4.05 \pm 0.40 \text{ Å}$ with 12.6% of frames sampling conformations beyond 5Å RMSD, indicating significant flexibility. In contrast, the R702W mutant showed reduced RMSD ($3.27 \pm 0.15 \text{ Å}$) with 0% of frames exceeding 5Å, demonstrating that the mutation rigidifies the protein structure (Figure 4). This reduced conformational sampling may impair the protein's ability to recognize and bind ligands.

Figure 4. RMSD distributions (A) and conformational heterogeneity (B) for WT vs R702W NOD2.

Lead Compound Identification

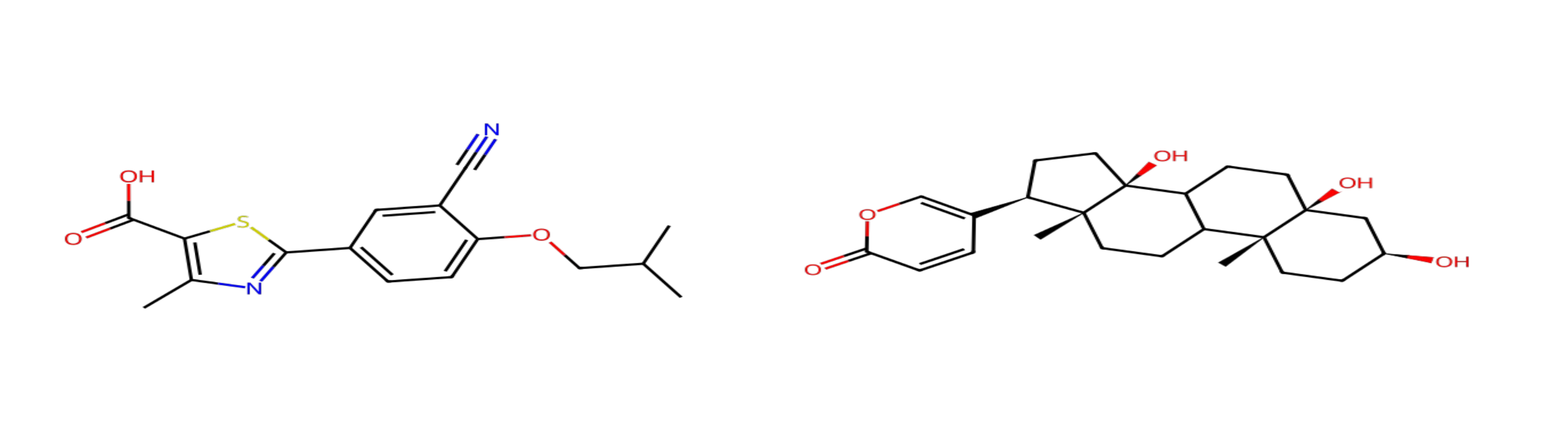


Figure 5. Chemical structures of Febuxostat (left) and Bufadienolide (right).

The computational pipeline identified two lead compounds from distinct chemical classes: Febuxostat, an FDA-approved xanthine oxidase inhibitor used to treat gout, and Bufadienolide (CID 10120), a cardiac glycoside-related natural product. Both compounds demonstrated stable binding pocket occupancy (70-80%) throughout MD simulations.

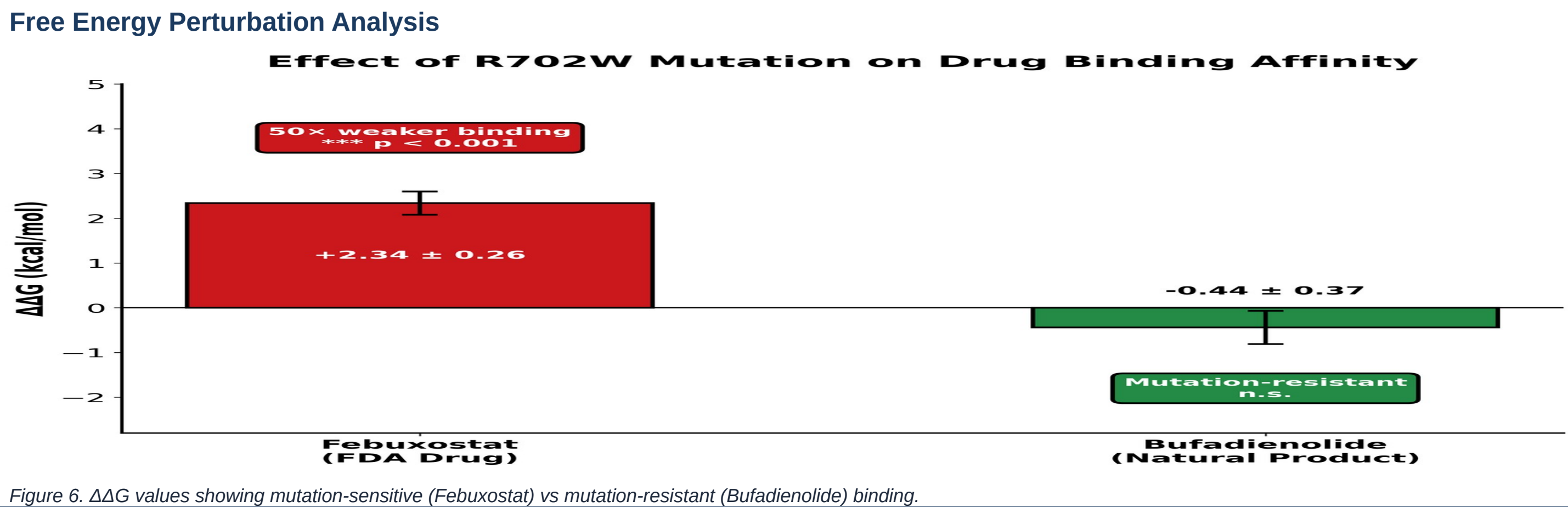


Figure 6. $\Delta\Delta G$ values showing mutation-sensitive (Febuxostat) vs mutation-resistant (Bufadienolide) binding.

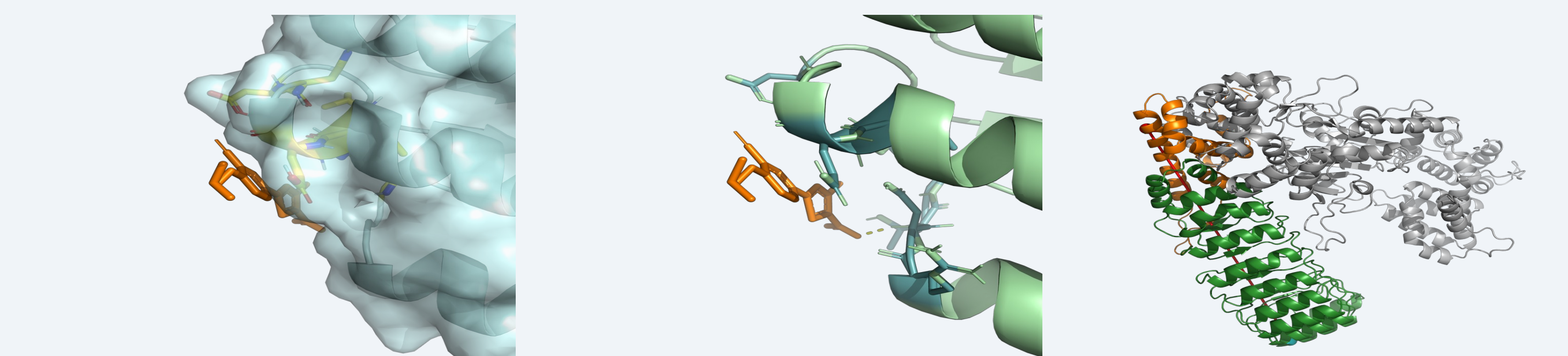


Figure 8. Binding pocket surface (left), ligand-protein interactions (center), and 79.4Å allosteric distance (right).

Results (continued)

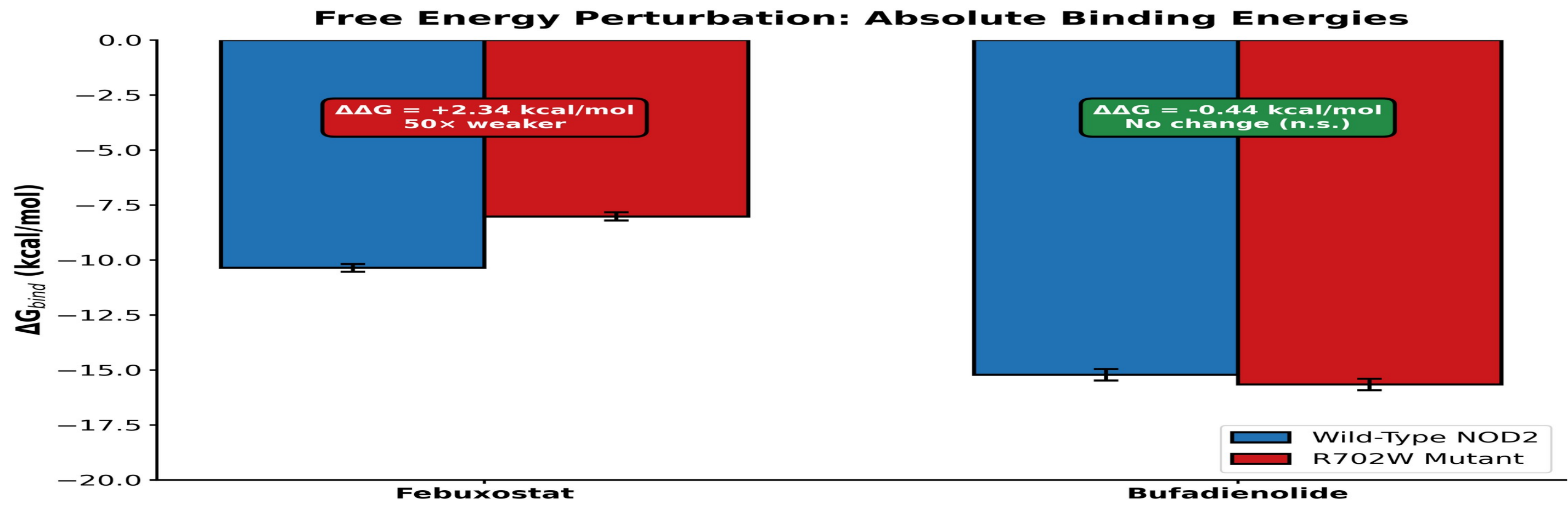


Figure 7. Absolute binding free energies for wild-type and R702W NOD2.

FEP calculations revealed dramatically different mutation sensitivities between the two compounds. Febuxostat exhibited a $\Delta\Delta G$ of $+2.34 \pm 0.26 \text{ kcal/mol}$ (8σ significance, $p < 0.001$), corresponding to approximately 50-fold weaker binding in the R702W mutant. Conversely, Bufadienolide showed a $\Delta\Delta G$ of $-0.44 \pm 0.37 \text{ kcal/mol}$ (1.8σ, not significant), indicating mutation-resistant binding.

Conclusion and Discussion

This study presents the first computational pipeline for identifying mutation-sensitive NOD2 modulators, integrating deep learning-based virtual screening with rigorous free energy calculations. The key findings are:

First, the R702W mutation induces protein rigidification, reducing conformational sampling as evidenced by decreased RMSD (3.27 Å vs 4.05 Å) and eliminated heterogeneity (0% vs 12.6% frames $> 5 \text{ Å}$). This reduced flexibility likely represents the mechanistic basis for impaired ligand recognition in mutation carriers.

Second, Febuxostat demonstrates mutation-sensitive binding with 50-fold weaker affinity for R702W NOD2 ($\Delta\Delta G = +2.34 \text{ kcal/mol}$, $p < 0.001$). This finding has important implications: Febuxostat-based therapeutics would be less effective in patients carrying this common Crohn's-associated variant.

Third, Bufadienolide exhibits mutation-resistant binding ($\Delta\Delta G = -0.44 \text{ kcal/mol}$, n.s.), maintaining equivalent affinity regardless of genotype. Notably, no prior literature reports NOD2-Bufadienolide interactions, establishing this as a novel therapeutic lead.

Fourth, the 79.4Å distance between the R702W mutation and the binding pocket confirms an allosteric mechanism whereby the mutation modulates binding through long-range conformational effects rather than direct steric interference.

Future Work

Experimental validation will be pursued using surface plasmon resonance (SPR) and isothermal titration calorimetry (ITC) to confirm binding affinities, with hydrogen-deuterium exchange mass spectrometry (HDX-MS) to map conformational changes upon ligand binding. The computational pipeline will be extended to characterize other Crohn's-associated NOD2 variants, including G908R and 1007fs. Structure-activity relationship (SAR) studies on Bufadienolide analogs may identify derivatives with improved drug-like properties while maintaining mutation-resistant binding. Finally, cell-based functional assays will assess whether identified compounds modulate NOD2-dependent NF-κB signaling.

Key References

- Hugot JP, Chamaillard M, Zouali H, et al. Association of NOD2 leucine-rich repeat variants with susceptibility to Crohn's disease. *Nature*. 2001;411:599-603.
- Jumper J, Evans R, Pritzel A, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*. 2021;596:583-589.
- McNutt AT, Francoeur P, Aggarwal R, et al. GNINA 1.0: molecular docking with deep learning. *J Cheminform*. 2021;13:43.
- Gapsys V, Pérez-Benito L, Aldeghi M, et al. Large scale relative protein ligand binding affinities using non-equilibrium alchemy. *Nat Protoc*. 2022;17:1114-1141.
- Abraham MJ, Murtola T, Schulz R, et al. GROMACS: High performance molecular simulations. *SoftwareX*. 2015;1-2:19-25.